Data Flow Diagram (DFD)

1. Level 0 DFD (System Context Diagram)

The Context Diagram represents the boundaries of the Opti-Crop system, showing inputs from users and outputs generated by the application.

```
+-------------------------------------------+
 |                                             |
 |                   FARMER /                  |
 |                  RESEARCHER                 |
 |                                             |
+------|---------------------------^-------+
 |                              |
 | 1. Inputs Soil/Weather Data   | 2. Receives Crop Recommendation
 |    (N, P, K, pH, Temp, etc.) |    & Tailored Soil Advice
v                  |
+------|---------------------------|-------+
 |                                             |
 |                 OPTI-CROP                   |
 |                APPLICATION                  |
 |                                             |
+-------------------------------------------+
```

2. Level 1 DFD (Process Level Diagram)

The Level 1 DFD details how data flows between UI processes, the Machine Learning inference engine, and the SQLite storage database.

1. **Process 1.0: User Management**
   • User inputs credentials (Register/Login).
   • Verifies credentials against the `users` table in `opticrop.db`.
   • Establishes user session.

2. **Process 2.0: ML Prediction Engine**
   • Receives soil features vector (1x7 array).
   • Loads trained model (`model.pkl`).

• Executes prediction -> outputs crop label.

3. **Process 3.0: Advice & Analytics Compiler**
   • Compares user inputs with optimal parameters loaded from `crop_ranges.pkl`.
   • Queries `crops` database table for water and seasonal details.
   • Computes nutrient deviations and drafts chemical/organic remediation advice.

4. **Process 4.0: Log Storage & History**
   • Saves inputs to `soil_data` table.
   • Saves prediction to `predictions` table.
   • Saves summary to `reports` table.
   • Updates user dashboard history.